Exercise 1. The rejection regions C_α for UMP hypothesis tests of significance level at most $\alpha \in (0,1)$ are often nested in the sense that $C_\alpha \subseteq C_{\alpha'}$ for all $0 < \alpha < \alpha' < 1$. This exercise demonstrates an example of UMP tests where this nesting behavior does not occur.

Let $\theta_0, \theta_1 \in \mathbb{R}$ be unequal parameters. Let H_0 denote the hypothesis $\{\theta = \theta_0\}$ and let H_1 denote the hypothesis $\{\theta = \theta_1\}$. Suppose $X \in \{1,2,3\}$ is a random variable. If $\theta = \theta_0$, assume that X takes the values $1,2,3$ with probabilities $.85, .1, .05$, respectively. If $\theta = \theta_1$, assume that X takes the values $1,2,3$ with probabilities $.7, .2, .1$, respectively. Let $\mathcal{T}$ denote the set of hypothesis tests with significance level at most α.

- Let $0 < \alpha < .15$. Show that a UMP class $\mathcal{T}$ test is not unique.
- When $\alpha = .05$, show there is a unique nonrandomized hypothesis UMP class $\mathcal{T}$ test.
- When $\alpha = .1$, show there is a unique nonrandomized hypothesis UMP class $\mathcal{T}$ test.
- Show that the $\alpha = .05$ and $\alpha' = .1$ UMP nonrandomized tests from above do not have nested rejection regions.
- However, when $\alpha = .05$ and $\alpha' = .1$, there are randomized UMP tests $\phi, \phi' \colon \mathbb{R}^n \to [0,1]$ respectively, that are nested in the sense that $\phi \leq \phi'$.

Solution: a) To find the Uniformly Most Powerful (UMP) test, we use the Neyman-Pearson Lemma. We compute the likelihood ratio $\Lambda(x) = \frac{P(X=x|\theta_1)}{P(X=x|\theta_0)}$ for each possible outcome of X:

- For $X = 1$: $\Lambda(1) = \frac{0.70}{0.85} = \frac{14}{17} \approx 0.824$
- For $X = 2$: $\Lambda(2) = \frac{0.20}{0.10} = 2.000$
- For $X = 3$: $\Lambda(3) = \frac{0.10}{0.05} = 2.000$

The likelihood ratios are ordered as: $\Lambda(1) < \Lambda(2) = \Lambda(3)$.
By the Neyman-Pearson Lemma, a test $\phi(x)$ maximizes power if it rejects H_0 when $\Lambda(x) > c$, accepts H_0 when $\Lambda(x) < c$, and optionally randomizes when $\Lambda(x) = c$.

To get a significance level $\alpha < 0.15$, we cannot reject $X = 1$ because $P(X = 1 \mid \theta_0) = 0.85$, leaving a maximum available size of $1 - 0.85 = 0.15$. Therefore, we must set $\phi(1) = 0$.

The critical threshold must be $c = 2$, where $\Lambda(2) = \Lambda(3) = 2$. Any test maximizing power must satisfy

$$\alpha = E_{\theta_0}[\phi(X)] = \phi(2)P(X = 2 \mid \theta_0) + \phi(3)P(X = 3 \mid \theta_0) = 0.10\phi(2) + 0.05\phi(3)$$

Since we have one constraint and two variables ($\phi(2), \phi(3) \in [0,1]$), there are **infinitely many combinations** of $\phi(2)$ and $\phi(3)$ that satisfy this equation for a given $\alpha < 0.15$. Every such combination yields the exact same maximal power:

$$\text{Power} = 0.20\phi(2) + 0.10\phi(3) = 2(0.10\phi(2) + 0.05\phi(3)) = 2\alpha$$

Because multiple distinct tests achieve this identical maximum power the UMP test is **not unique**.

b) For a nonrandomized test, $\phi(x)$ must be either 0 or 1. We need:

$$0.10\phi(2) + 0.05\phi(3) \leq 0.05$$

To maximize power $(0.20\phi(2) + 0.10\phi(3))$ under this constraint with binary values:

- If $\phi(2) = 1$, then $0.10(1) + 0.05\phi(3) \leq 0.05 \implies 0.10 \leq 0.05$, which is impossible. Thus, $\phi(2)$ must be 0.
- If $\phi(2) = 0$, we can set $\phi(3) = 1$, which gives an exact size of $0.05 \leq 0.05$ and a power of 0.10.

Thus, the unique nonrandomized UMP test for $\alpha = 0.05$ has the rejection region

$$C_{0.05} = \{3\}$$

c) We require a nonrandomized test satisfying

$$0.10\phi(2) + 0.05\phi(3) \leq 0.10$$

Let's evaluate the options to maximize power $(0.20\phi(2) + 0.10\phi(3))$:

- Option A: Set $\phi(2) = 1, \phi(3) = 0$. Size $= 0.10 \leq 0.10$. Power $= 0.20$.
- Option B: Set $\phi(2) = 0, \phi(3) = 1$. Size $= 0.05 \leq 0.10$. Power $= 0.10$.

Option A yields strictly higher power than Option B. Thus, the unique nonrandomized UMP test for $\alpha = 0.1$ has the rejection region:

$$C_{0.1} = \{2\}$$

d) Comparing our two unique nonrandomized rejection regions:

- $C_{0.05} = \{3\}$
- $C_{0.1} = \{2\}$

For nesting to hold, we would require $C_{0.05} \subseteq C_{0.1}$. However $\{3\} \not\subseteq \{2\}$ so the rejection regions are completely disjoint, proving that the nonrandomized UMP tests are not nested.

e) We try to construct a pair of tests ϕ and ϕ' such that $\phi \leq \phi'$ everywhere:

1. For $\alpha = 0.05$: Choose the nonrandomized UMP test we found earlier:

$$\phi(1) = 0, \quad \phi(2) = 0, \quad \phi(3) = 1 \quad (\text{Size} = 0.05)$$

2. For $\alpha' = 0.1$: We can construct a randomized UMP test by keeping $\phi'(3) = 1$ and solving for $\phi'(2)$ using the size equation:

$$0.10\phi'(2) + 0.05(1) = 0.10 \implies 0.10\phi'(2) = 0.05 \implies \phi'(2) = 0.5$$

This gives the randomized test:

$$\phi'(1) = 0, \quad \phi'(2) = 0.5, \quad \phi'(3) = 1 \quad (\text{Size} = 0.10)$$

Checking the nesting condition pointwise we have

- $X = 1$**:** $0 \leq 0$ (True)
- $X = 2$**:** $0 \leq 0.5$ (True)
- $X = 3$**:** $1 \leq 1$ (True)

Since both are valid UMP tests for their respective levels and $\phi \leq \phi'$ across all outcomes nested randomized UMP tests do exist.

Exercise 2. Suppose X is a Gaussian distributed random variable with known variance $\sigma^2 > 0$ but unknown mean. Fix $\mu_0, \mu_1 \in \mathbb{R}$. Assume that $\mu_0 - \mu_1 > 0$. We want to test the hypothesis H_0 that $\mu = \mu_0$ versus the hypothesis H_1 that $\mu = \mu_1$. Fix $\alpha \in (0, 1)$. Explicitly describe the UMP test for the class of tests whose significance level is at most α.

Your description of the test should use the function $\Phi(t) := \int_{-\infty}^{t} e^{-x^2/2} dx/\sqrt{2\pi}$, $\Phi\colon \mathbb{R} \to (0, 1)$, and/or the function $\Phi^{-1}\colon (0, 1) \to \mathbb{R}$. (Recall that $\Phi(\Phi^{-1}(s)) = s$ for all $s \in (0, 1)$ and $\Phi^{-1}(\Phi(t)) = t$ for all $t \in \mathbb{R}$.)

Solution: We have that the Uniformly Most Powerful (UMP) test at significance level α rejects H_0 in favor of H_1 if $X < \mu_0 + \sigma\Phi^{-1}(\alpha)$.

By Neyman-Pearson Lemma the most powerful test rejects H_0 when the likelihood ratio $\Lambda(x) = \frac{f(x|\mu_1)}{f(x|\mu_0)}$ is strictly greater than a threshold c.

The probability density function for a Gaussian random variable $X \sim N(\mu, \sigma^2)$ is

$$f(x \mid \mu) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Plugging this into the likelihood ratio gives:

$$\Lambda(x) = \frac{\exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right)}{\exp\left(-\frac{(x-\mu_0)^2}{2\sigma^2}\right)} = \exp\left(\frac{(x-\mu_0)^2 - (x-\mu_1)^2}{2\sigma^2}\right)$$

Expanding the exponents inside the expression yields:

$$(x-\mu_0)^2 - (x-\mu_1)^2 = 2x(\mu_1 - \mu_0) + \mu_0^2 - \mu_1^2$$

The test rejects H_0 when $\Lambda(x) > c$. Taking the natural logarithm of both sides:

$$\frac{2x(\mu_1 - \mu_0) + \mu_0^2 - \mu_1^2}{2\sigma^2} > \ln c$$

$$2x(\mu_1 - \mu_0) > 2\sigma^2 \ln c - \mu_0^2 + \mu_1^2$$

We are given that $\mu_0 - \mu_1 > 0$, meaning $\mu_1 - \mu_0 < 0$. Dividing by a negative number flips the direction of the inequality

$$x < \frac{2\sigma^2 \ln c - \mu_0^2 + \mu_1^2}{2(\mu_1 - \mu_0)}$$

Thus, the rejection region simplifies to the form $X < k$ for some constant k.

To satisfy the significance level α, we set the probability of a Type I error (rejecting H_0 when H_0 is true) exactly equal to α:

$$P_{\mu_0}(X < k) = \alpha$$

Under H_0, $X \sim N(\mu_0, \sigma^2)$, so the standardized variable $Z = \frac{X-\mu_0}{\sigma}$ follows a standard normal distribution $N(0, 1)$.

$$P_{\mu_0}\left(\frac{X-\mu_0}{\sigma} < \frac{k-\mu_0}{\sigma}\right) = \alpha$$

$$\Phi\left(\frac{k-\mu_0}{\sigma}\right) = \alpha$$

Applying the inverse standard normal CDF Φ^{-1} to both sides:

$$\frac{k-\mu_0}{\sigma} = \Phi^{-1}(\alpha) \implies k = \mu_0 + \sigma\Phi^{-1}(\alpha)$$

The UMP test function $\phi(X)$ is defined as:

$$\phi(X) = \begin{cases} 1 & \text{if } X < \mu_0 + \sigma\Phi^{-1}(\alpha) \\ 0 & \text{if } X \geq \mu_0 + \sigma\Phi^{-1}(\alpha) \end{cases}$$

Where 1 means reject H_0 and 0 means fail to reject H_0.

Exercise 3. This exercise demonstrates that a UMP might not always exists.

Let $X_1, \ldots, X_n$ be i.i.d. Gaussian random variables with known variance and unknown mean $\mu \in \mathbb{R}$. Fix $\mu_0 \in \mathbb{R}$. Let H_0 denote the hypothesis $\{\mu = \mu_0\}$ and let H_1 denote the hypothesis $\mu \neq \mu_0$. Fix $0 < \alpha < 1$. Let $\mathcal{T}$ denote the set of hypothesis tests with significance level at most α. Show that no UMP class $\mathcal{T}$ test exists, using the following strategy.

- Let $\mu_1 < \mu_0$. You may take as given the following fact (that follows from the Karlin-Rubin Theorem): the power at μ_1 is maximized among class $\mathcal{T}$ tests by the hypothesis test ϕ that rejects H_0 when the sample mean satisfies $\overline{X} < c$ for an appropriate choice of $c \in \mathbb{R}$. Assume for the sake of contradiction that a UMP class $\mathcal{T}$ test ϕ' exists. Then, using the necessity part of the Neyman-Pearson Lemma (i.e. consider testing $\mu = \mu_0$ versus $\mu = \mu_1$), conclude that ϕ' must have the same rejection region as ϕ (just by examining the power of the tests at μ_1.)
- Consider now a test in $\mathcal{T}$ that rejects H_0 when the sample mean satisfies $\overline{X} > c'$ for an appropriate choice of $c' \in \mathbb{R}$. Repeating the previous argument, conclude that ϕ' must reject when $\overline{X} > c'$, leading to a contradiction.

  That is, let $\mu_2 > \mu_0$. You may take as given the following fact (that follows from the Karlin-Rubin Theorem): the power at μ_2 is maximized among class $\mathcal{T}$ tests by the hypothesis test ϕ'' that rejects H_0 when the sample mean satisfies $\overline{X} > c'$ for an appropriate choice of $c' \in \mathbb{R}$. Then, using the necessity part of the Neyman-Pearson Lemma (i.e. consider testing $\mu = \mu_0$ versus $\mu = \mu_2$), conclude that ϕ' must have the same rejection region as ϕ''.

Solution: Assume by contradiction that a Uniformly Most Powerful (UMP) class $\mathcal{T}$ test, denoted by ϕ', exists. By definition, a UMP test must maximize the power function at every parameter value in the alternative hypothesis space $\mu \in H_1$.

Let $\mu_1 < \mu_0$. By the Karlin-Rubin Theorem (or the monotonicity of the likelihood ratio for the normal distribution), the power at μ_1 is uniquely maximized among all level-α tests by the hypothesis test ϕ that rejects H_0 when the sample mean satisfies $\bar{X} < c$ for an appropriate choice of $c \in \mathbb{R}$.

Since ϕ' is assumed to be a UMP test, its power at μ_1 must be at least as large as the power of any other class $\mathcal{T}$ test. Thus, we have

$$\beta_{\phi'}(\mu_1) \geq \beta_{\phi}(\mu_1)$$

Since ϕ already achieves the maximum possible power at μ_1, it must be that $\beta_{\phi'}(\mu_1) = \beta_\phi(\mu_1)$.

By the necessity part of the Neyman-Pearson Lemma applied to the simple versus simple testing problem $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1$, any test that achieves the maximum power must have the same rejection region as the likelihood ratio test (except possibly on a set of Lebesgue measure zero). Therefore, ϕ' must have the same rejection region as ϕ:

$$\mathcal{R}' = \{\bar{X} < c\} \quad \text{almost everywhere.} \tag{1}$$

Now let $\mu_2 > \mu_0$. By a symmetric application of the Karlin-Rubin Theorem, the power at μ_2 is uniquely maximized among all level-α tests by the hypothesis test ϕ'' that rejects H_0 when the sample mean satisfies $\bar{X} > c'$ for an appropriate choice of $c' \in \mathbb{R}$.

Since ϕ' is a UMP test, it must also maximize the power at μ_2. By following the exact same logical steps as before, the power of ϕ' at μ_2 must equal the maximum possible power achieved by ϕ'':

$$\beta_{\phi'}(\mu_2) = \beta_{\phi''}(\mu_2)$$

Applying the necessity part of the Neyman-Pearson Lemma to the simple versus simple testing problem $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_2$, we conclude that ϕ' must have the same rejection region as ϕ'':

$$\mathcal{R}' = \{\bar{X} > c'\} \quad \text{almost everywhere.} \tag{2}$$

From equations (1) and (2), the rejection region $\mathcal{R}'$ of our assumed UMP test ϕ' must satisfy

$$\{\bar{X} < c\} = \{\bar{X} > c'\} \quad \text{almost everywhere.}$$

Since $0 < \alpha < 1$, both c and c' are finite real constants. The set of sample outcomes where $\bar{X} < c$ is fundamentally disjoint from (or cannot be identical to) the set where $\bar{X} > c'$ except on a trivial empty or measure-zero intersection.

Because a rejection region cannot simultaneously be a strictly left-tailed event and a strictly right-tailed event for a continuous random variable, we have reached a logical contradiction.

Exercise 4. Prove the following version of the Karlin-Rubin Theorem, with the inequalities reversed in the definition of the hypotheses.

Let $\{f_\theta\}$ be a family of PDFs with the MLR property, with respect to a real-valued statistic $Y = t(X)$, where $\theta \in \Theta \subseteq \mathbb{R}$. Let $0 \leq \gamma \leq 1$. Fix $\theta_0 \in \Theta$. Consider the hypothesis $H_0 = \{\theta \geq \theta_0\}$ and the hypothesis $H_1 = \{\theta < \theta_0\}$. Let $c \in \mathbb{R}$. Consider the randomized hypothesis test $\phi\colon \mathbb{R}^n \to [0, 1]$ defined by

$$\phi(x) := \begin{cases} 0, & \text{if } t(x) > c \\ 1, & \text{if } t(x) < c \\ \gamma, & \text{if } t(x) = c. \end{cases}$$

Define $\alpha := E_{\theta_0}\phi(X)$. Let $\mathcal{T}$ be the class of all randomized hypothesis tests with significance level at most α.

(i) ϕ is UMP class $\mathcal{T}$.

(iii) β, the power function of ϕ, is nonincreasing and strictly decreasing when it takes values in $(0, 1)$.

Solution: By definition, the family $\{f_\theta\}$ has the Monotone Likelihood Ratio (MLR) property in $Y = t(X)$ if for any $\theta_1 < \theta_2$, the likelihood ratio $f_{\theta_2}(x)/f_{\theta_1}(x)$ is a nondecreasing function of $t(x)$.

Let $\beta(\theta) = E_\theta\phi(X)$ be the power function of the test ϕ. We want to show that $\beta(\theta)$ is nonincreasing, and strictly decreasing when $\beta(\theta) \in (0, 1)$.

Fix any $\theta_1, \theta_2 \in \Theta$ such that $\theta_1 < \theta_2$. Because the family has MLR in $t(X)$, the ratio $f_{\theta_2}(x)/f_{\theta_1}(x)$ is nondecreasing in $t(x)$.

The test $\phi(x)$ is defined as:

$$\phi(x) = \begin{cases} 1 & \text{if } t(x) < c \\ \gamma & \text{if } t(x) = c \\ 0 & \text{if } t(x) > c \end{cases}$$

Notice that $\phi(x)$ is a nonincreasing function of $t(x)$. Because $f_{\theta_2}(x)/f_{\theta_1}(x)$ is nondecreasing in $t(x)$ and $\phi(x)$ is nonincreasing in $t(x)$, they move in opposite directions. Consequently, for any x_1, x_2 such that $t(x_1)k \implies \frac{f_{\theta_0}(x)}{f_{\theta_1}(x)} < \frac{1}{k}$. Because $\{f_0\}$ has the MLR property in $t(X)$ and $\theta_1 < \theta_0$, the ratio $f_{\theta_0}(x)/f_{\theta_1}(x)$ is a nondecreasing function of $t(x)$. Therefore, the condition $f_{\theta_0}(x)/f_{\theta_1}(x) < 1/k$ is exactly equivalent to $t(x) < c$ for some constant c.

The Neyman-Pearson test for H_0' vs H_1' is precisely:

$$\phi(x) = \begin{cases} 1 & \text{if } t(x) < c \\ \gamma & \text{if } t(x) = c \\ 0 & \text{if } t(x) > c \end{cases}$$

where c and γ are chosen such that $E_{\theta_0}\phi(X) = \alpha$.

By the Neyman-Pearson Lemma, ϕ maximizes the power at $\theta = \theta_1$ among all tests ψ satisfying $E_{\theta_0}\psi(X) \leq \alpha$.

Since any test $\psi \in T$ must satisfy $\sup_{\theta \geq \theta_0} E_\theta\psi(X) \leq \alpha$, it must specifically satisfy $E_{\theta_0}\psi(X) \leq \alpha$. Therefore:

$$E_{\theta_1}\phi(X) \geq E_{\theta_1}\psi(X)$$

Since $\theta_1 < \theta_0$ was chosen arbitrarily, ϕ maximizes power at every single point in H_1. Thus, ϕ is UMP within the class T.

Exercise 5. Prove the following one-sided version of the Karlin-Rubin Theorem.

Let $\{f_\theta\}$ be a family of PDFs with the MLR property, with respect to a real-valued statistic $Y = t(X)$, where $\theta \in \Theta \subseteq \mathbb{R}$. Let $0 \leq \gamma \leq 1$. Fix $\theta_0 \in \Theta$. Consider the hypothesis $H_0 = \{\theta = \theta_0\}$ and the hypothesis $H_1 = \{\theta > \theta_0\}$. Let $c \in \mathbb{R}$. Consider the randomized hypothesis test $\phi\colon \mathbb{R}^n \to [0, 1]$ defined by

$$\phi(x) := \begin{cases} 1, & \text{if } t(x) > c \\ 0, & \text{if } t(x) < c \\ \gamma, & \text{if } t(x) = c. \end{cases}$$

Define $\alpha := E_{\theta_0}\phi(X)$. Let $\mathcal{T}$ be the class of all randomized hypothesis tests with significance level at most α.

Then ϕ is UMP class $\mathcal{T}$.

Solution: To prove this one-sided version of the Karlin-Rubin Theorem we apply the fundamental Neyman-Pearson Lemma. The core strategy is to demonstrate that the given test $\phi(x)$

maximizes the power against any specific alternative $\theta_1 > \theta_0$. Because the critical region of the test does not depend on the specific choice of θ_1, it is Uniformly Most Powerful for the composite alternative hypothesis $H_1 : \theta > \theta_0$.

Fix an arbitrary parameter value $\theta_1 \in \Theta$ such that $\boldsymbol{\theta_1 > \theta_0}$. Consider the simplified simple-vs-simple hypothesis testing problem:

$$H_0' : \theta = \theta_0$$
$$H_1' : \theta = \theta_1$$

By assumption, the family of densities $\{f_\theta\}$ possesses the Monotone Likelihood Ratio property in the real-valued statistic $Y = t(X)$.

This implies that for our chosen $\theta_1 > \theta_0$, the likelihood ratio is a non-decreasing function of $t(x)$:

$$\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = g(t(x))$$

where $g(y)$ is a non-decreasing function of y.

According to the Neyman-Pearson Lemma, the most powerful test $\phi^*(x)$ at significance level α for testing H_0' against H_1' has the form:

$$\phi^*(x) = \begin{cases} 1 & \text{if } \frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} > k \\ 0 & \text{if } \frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} < k \\ \gamma^* & \text{if } \frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} = k \end{cases}$$

where $k \geq 0$ and $\gamma^* \in [0, 1]$ are constants chosen to satisfy the size constraint $E_{\theta_0}\phi^*(X) = \alpha$.

Because $g(y) = \frac{f_{\theta_1}(x)}{f_{\theta_0}(x)}$ is non-decreasing in $t(x)$, the condition $\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)} > k$ is structurally equivalent to $\mathbf{t}(\mathbf{x}) > \mathbf{c}$ for some constant $c \in \mathbb{R}$ (except possibly on sets of probability zero under both hypotheses).

Thus, the Neyman-Pearson most powerful test can be rewritten directly in terms of the statistic $t(x)$ as:

$$\phi(x) = \begin{cases} 1 & \text{if } t(x) > c \\ 0 & \text{if } t(x) < c \\ \gamma & \text{if } t(x) = c \end{cases}$$

where c and γ are uniquely determined by the size constraint:

$$E_{\theta_0}\phi(X) = P_{\theta_0}(t(X) > c) + \gamma P_{\theta_0}(t(X) = c) = \alpha$$

Let $\mathcal{T}$ be the class of all randomized tests satisfying $E_{\theta_0}\psi(X) \leq \alpha$.

1. By the Neyman-Pearson Lemma, the test $\phi(x)$ maximizes power against the specific alternative θ_1. That is:
$$E_{\theta_1}\phi(X) \geq E_{\theta_1}\psi(X) \quad \forall \psi \in \mathcal{T}$$

2. Notice the crucial point of independence: The definition of the test $\phi(x)$ depends only on c and γ, which are determined entirely by θ_0 and α. The structure of $\phi(x)$ does not change regardless of which specific value of $\theta_1 > \theta_0$ we select.

Since $\phi \in \mathcal{T}$ and ϕ maximizes the power function $E_{\theta_1}\phi(X)$ simultaneously for every single choice of $\theta_1 > \theta_0$, ϕ is the Uniformly Most Powerful (UMP) test of size α for $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$ within the class $\mathcal{T}$.

Exercise 6. Let $X_1, \ldots, X_n$ be i.i.d. random variables. Let $X = (X_1, \ldots, X_n)$. Let $\theta > 0$. Assume that X_1 is uniformly distributed in the interval $[0, \theta]$. Fix $\theta_0 > 0$. Fix $0 < \alpha < 1$. Let $\mathcal{T}$ denote the set of hypothesis tests with significance level at most α.

- Suppose we test $H_0 = \{\theta \leq \theta_0\}$ versus $H_1 = \{\theta > \theta_0\}$. Identify the set of all UMP class $\mathcal{T}$ hypothesis tests.
- Suppose we test $H_0 = \{\theta = \theta_0\}$ versus $H_1 = \{\theta \neq \theta_0\}$. Show there is a unique UMP class $\mathcal{T}$ hypothesis test in this case.

(Hint: first consider testing $\{\theta = \theta_0\}$ versus $\{\theta = \theta_1\}$ with $\theta_1 > \theta_0$, and apply the Neyman-Pearson Lemma. That is, mimic the argument of the Karlin-Rubin Theorem.) (As an aside, observe that, if you naïvely apply the Karlin-Rubin Theorem, you will not find all UMP tests, i.e. a non-strict MLR property version of the Karlin-Rubin Theorem will neglect some UMP tests.)

Solution: Let $X = (X_1, \ldots, X_n)$ be i.i.d. random variables distributed uniformly on the interval $[0, \theta]$. The joint probability density function of X is given by:

$$f(x; \theta) = \frac{1}{\theta^n} \mathbb{I}_{\{0 \leq X_{(n)} \leq \theta\}} \mathbb{I}_{\{X_{(1)} \geq 0\}}$$

where $X_{(n)} = \max(X_1, \ldots, X_n)$ is the sufficient statistic for θ.

To find all Uniformly Most Powerful (UMP) tests in class $\mathcal{T}$ (significance level at most α), we seek a test function $\phi(X)$ that maximizes the power $\beta(\theta) = \mathbb{E}_\theta[\phi(X)]$ for all $\theta > \theta_0$, subject to the size constraint $\mathbb{E}_\theta[\phi(X)] \leq \alpha$ for all $\theta \leq \theta_0$.

For any alternative value $\theta > \theta_0$, if $X_{(n)} > \theta_0$, the sample has probability zero under H_0. To maximize power without affecting the size under H_0, any UMP test must reject with probability 1 when $X_{(n)} > \theta_0$.

The power function for $\theta > \theta_0$ can be expanded as:

$$\beta(\theta) = \mathbb{P}_\theta(X_{(n)} > \theta_0) + \mathbb{E}_\theta \left[\phi(X) \mathbb{I}_{\{X_{(n)} \leq \theta_0\}} \right]$$

Since $\mathbb{P}_\theta(X_{(n)} > \theta_0) = 1 - \left(\frac{\theta_0}{\theta}\right)^n$, maximizing power reduces to maximizing the second term. Under the condition $X_{(n)} \leq \theta_0$, the conditional distribution of X does not depend on θ for any $\theta \geq \theta_0$. Thus, maximizing this term for any $\theta > \theta_0$ is equivalent to maximizing it at $\theta = \theta_0$, where the upper bound allowed by the size constraint is exactly α.

Therefore, the set of all UMP class $\mathcal{T}$ hypothesis tests consists of all test functions $\phi(X)$ satisfying:

1. $\phi(X) = 1$ when $X_{(n)} > \theta_0$.
2. $\mathbb{E}_{\theta_0} \left[\phi(X) \mathbb{I}_{\{X_{(n)} \leq \theta_0\}} \right] = \alpha$.

Any test of the following form belongs to the UMP class:

$$\phi(X) = \begin{cases} 1 & \text{if } X_{(n)} > \theta_0 \\ \gamma(X) & \text{if } X_{(n)} \leq \theta_0 \end{cases}$$

where $\gamma(X) \in [0,1]$ is any critical function satisfying $\mathbb{E}_{\theta_0}[\gamma(X)] = \alpha$ over the region $X_{(n)} \leq \theta_0$.

To analyze the two-sided case, we apply the Neyman-Pearson Lemma to simple directional alternatives.

Consider a simple alternative $\theta_1 > \theta_0$. The likelihood ratio is:

$$\Lambda(X) = \frac{f(x;\theta_1)}{f(x;\theta_0)} = \frac{\frac{1}{\theta_1^n}\mathbb{I}_{\{X_{(n)} \leq \theta_1\}}}{\frac{1}{\theta_0^n}\mathbb{I}_{\{X_{(n)} \leq \theta_0\}}}$$

- If $X_{(n)} > \theta_0$, the denominator is 0, meaning $\Lambda(X) = \infty$, so the test must reject.
- If $X_{(n)} \leq \theta_0$, the ratio is a constant $\left(\frac{\theta_0}{\theta_1}\right)^n$.

To achieve size α at $\theta = \theta_0$, the unique Neyman-Pearson test is:

$$\phi_1(X) = \begin{cases} 1 & \text{if } X_{(n)} > \theta_0 \\ \alpha & \text{if } X_{(n)} \leq \theta_0 \end{cases}$$

Because this test structure does not depend on the specific choice of θ_1, $\phi_1(X)$ is the unique UMP test for $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$.

Now consider an alternative $\theta_2 < \theta_0$. The likelihood ratio is:

$$\Lambda(X) = \frac{\frac{1}{\theta_2^n}\mathbb{I}_{\{X_{(n)} \leq \theta_2\}}}{\frac{1}{\theta_0^n}\mathbb{I}_{\{X_{(n)} \leq \theta_0\}}}$$

- If $X_{(n)} > \theta_2$, $\Lambda(X) = 0$.
- If $X_{(n)} \leq \theta_2$, $\Lambda(X) = \left(\frac{\theta_0}{\theta_2}\right)^n > 1$.

The test maximizes power by rejecting for small values of $X_{(n)}$. Setting the size to α gives $\mathbb{P}_{\theta_0}(X_{(n)} \leq c) = \left(\frac{c}{\theta_0}\right)^n = \alpha$, which yields the critical threshold $c = \theta_0\alpha^{1/n}$. Thus, the unique UMP test for $H_0 : \theta = \theta_0$ against $H_1 : \theta < \theta_0$ is:

$$\phi_2(X) = \begin{cases} 1 & \text{if } X_{(n)} \leq \theta_0\alpha^{1/n} \\ 0 & \text{if } X_{(n)} > \theta_0\alpha^{1/n} \end{cases}$$

For a single test to be UMP against $H_1 : \theta \neq \theta_0$, it must simultaneously match the maximum power characteristics of both one-sided UMP tests:

- To be UMP against $\theta > \theta_0$, it must equal α when $X_{(n)} \leq \theta_0$.
- To be UMP against $\theta < \theta_0$, it must equal 1 when $X_{(n)} \leq \theta_0\alpha^{1/n}$.

Combining these requirements forces a unique structural candidate test $\phi^*(X)$:

$$\phi^*(X) = \begin{cases} 1 & \text{if } X_{(n)} > \theta_0 \\ \alpha & \text{if } \theta_0\alpha^{1/n} < X_{(n)} \leq \theta_0 \\ 1 & \text{if } X_{(n)} \leq \theta_0\alpha^{1/n} \end{cases}$$

Evaluating the expected size of $\phi^*(X)$ under H_0 ($\theta = \theta_0$):

$$\mathbb{E}_{\theta_0}[\phi^*(X)] = 0 + \alpha(1-\alpha) + \alpha = 2\alpha - \alpha^2$$

Since $2\alpha - \alpha^2 > \alpha$ for all $0 < \alpha < 1$, this candidate test violates the size requirement of class $\mathcal{T}$. Because the directional optimization parameters strictly conflict with one another under a single size budget α, a standard two-sided UMP test does not exist in this framework.

Exercise 7. Let $X_1, \ldots, X_n$ be i.i.d. random variables that are uniformly distributed in the interval $[\theta, \theta + 1]$, where $\theta \in \mathbb{R}$ is an unknown parameter. Fix $\theta_0 \in \mathbb{R}$. Suppose we want to test the hypothesis that $\theta \leq \theta_0$ versus $\theta > \theta_0$. For any $0 \leq \alpha \leq 1$, show that there exists a UMP test among tests with significance level at most α, and this test rejects the null hypothesis when $X_{(1)} > \theta_0 + c(\alpha)$ or $X_{(n)} > \theta_0 + 1$.

On the other hand, show that the joint density of $X_1, \ldots, X_n$ does not have the MLR property with respect to any statistic (when $n > 1$). (Hint: if it did have the MLR property, what would the Karlin-Rubin Theorem imply about the UMP rejection regions?)

(Hint: for the first part, first consider testing $\{\theta = \theta_0\}$ versus $\{\theta = \theta_1\}$ with $\theta_1 > \theta_0$, and apply the Neyman-Pearson Lemma.)

Solution: Let $X_1, \ldots, X_n$ be i.i.d. random variables uniformly distributed on the interval $[\theta, \theta + 1]$. The joint probability density function (pdf) of the sample $\mathbf{X} = (X_1, \ldots, X_n)$ can be expressed using indicator functions as:

$$f(\mathbf{x}; \theta) = \prod_{i=1}^{n} \mathbb{I}_{[\theta, \theta+1]}(x_i) = \mathbb{I}_{\{x_{(1)} \geq \theta\}} \mathbb{I}_{\{x_{(n)} \leq \theta+1\}}$$

where $x_{(1)} = \min(x_i)$ and $x_{(n)} = \max(x_i)$ are the smallest and largest order statistics, respectively.

Following the hint, we first consider testing the simple null hypothesis $H_0 : \theta = \theta_0$ against the simple alternative $H_1 : \theta = \theta_1$, where $\theta_1 > \theta_0$. By the Neyman-Pearson lemma, the most powerful test rejects H_0 when the likelihood ratio exceeds some threshold $k \geq 0$:

$$\Lambda(\mathbf{x}) = \frac{f(\mathbf{x}; \theta_1)}{f(\mathbf{x}; \theta_0)} = \frac{\mathbb{I}_{\{x_{(1)} \geq \theta_1\}} \mathbb{I}_{\{x_{(n)} \leq \theta_1+1\}}}{\mathbb{I}_{\{x_{(1)} \geq \theta_0\}} \mathbb{I}_{\{x_{(n)} \leq \theta_0+1\}}} > k$$

Let us evaluate the behavior of this ratio based on the observed sample space:

- If $x_{(n)} > \theta_0 + 1$, the denominator becomes 0. Since $\theta_1 > \theta_0$, the numerator can be non-zero (if $x_{(1)} \geq \theta_1$ and $x_{(n)} \leq \theta_1 + 1$). This yields a likelihood ratio of ∞. Therefore, any valid test must reject H_0 in this region.
- If $x_{(1)} < \theta_1$, the numerator becomes 0, yielding a likelihood ratio of 0. We fail to reject H_0 here.
- In the overlapping support region where both indicator conditions are met, the ratio evaluates to a constant value of 1.

Thus, the critical region that maximizes power consists of rejecting H_0 whenever $x_{(n)} > \theta_0 + 1$, or when $x_{(1)}$ is sufficiently large within the remaining support. This establishes the structural form of the rejection region:

$$\mathcal{R} = \{\mathbf{x} : x_{(1)} > \theta_0 + c(\alpha) \text{ or } x_{(n)} > \theta_0 + 1\}$$

We select $c(\alpha)$ to control the size of the test at exactly α under H_0. Note that when $\theta = \theta_0$, the event $\{x_{(n)} > \theta_0 + 1\}$ occurs with probability 0. Thus:

$$P_{\theta_0}(\mathcal{R}) = P_{\theta_0}(x_{(1)} > \theta_0 + c(\alpha)) = \alpha$$

Under $\theta = \theta_0$, each $X_i \sim U[\theta_0, \theta_0 + 1]$. The survival function of the minimum order statistic $X_{(1)}$ for $t \in [\theta_0, \theta_0 + 1]$ is given by:

$$P_{\theta_0}(X_{(1)} > t) = \prod_{i=1}^{n} P_{\theta_0}(X_i > t) = \left(\frac{\theta_0 + 1 - t}{1}\right)^n = (\theta_0 + 1 - t)^n$$

Setting $t = \theta_0 + c(\alpha)$ yields:

$$(\theta_0 + 1 - (\theta_0 + c(\alpha)))^n = (1 - c(\alpha))^n = \alpha \implies c(\alpha) = 1 - \alpha^{1/n}$$

Since this rejection region $\mathcal{R}$ depends only on θ_0 and α and remains identical for any chosen value of $\theta_1 > \theta_0$ it is the Uniformly Most Powerful test for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$.

We proceed by contradiction using the structural constraints of the Karlin-Rubin Theorem.

1. The Karlin-Rubin Requirement: If a family of densities possesses the Monotone Likelihood Ratio property with respect to a scalar statistic $T(\mathbf{X})$, the theorem guarantees that the UMP test for a one-sided hypothesis must have a decision boundary defined solely by that single statistic, taking the form:
$$\mathcal{R}_{KR} = \{\mathbf{x} : T(\mathbf{x}) > C\}$$

2. The Contradiction: As derived in Part 1, the unique UMP rejection region for this uniform distribution model is:
$$\mathcal{R} = \{\mathbf{x} : x_{(1)} > \theta_0 + c(\alpha) \text{ or } x_{(n)} > \theta_0 + 1\}$$
This region explicitly depends on two distinct statistics ($X_{(1)}$ and $X_{(n)}$) whose individual operational boundaries alter independently depending on the chosen value of θ_0.

3. Geometric Boundary Discrepancy: In the sample space, the boundary of $\mathcal{R}$ forms an L-shaped region. Because this boundary changes shape unevenly across different values of θ_0 and α, no single time-invariant scalar statistic $T(\mathbf{X})$ can perfectly map or simplify this multi-threshold boundary.

Since the true UMP test cannot be structurally formatted into a single-statistic threshold, the joint density does not possess the MLR property with respect to any statistic when $n > 1$.

Exercise 8. Let $\{f_\theta : \theta \in \mathbb{R}\}$ be a family of positive, single-variable PDFs, i.e. $f_\theta \colon \mathbb{R} \to (0, \infty)$ for all $\theta \in \mathbb{R}$. Assume that $f_\theta(x)$ is twice continuously differentiable in the parameters θ, x.

Show that $\{f_\theta\}$ has the MLR property with respect to the statistic $t(x) = x$ ($x \in \mathbb{R}$) if and only if

$$\frac{\partial^2}{\partial\theta\,\partial x} \log f_\theta(x) \geq 0, \qquad \forall\, x, \theta \in \mathbb{R}.$$

Exercise 8. Let $\{f_\theta : \theta \in \mathbb{R}\}$ be a family of positive, single-variable PDFs, i.e. $f_\theta : \mathbb{R} \to (0, \infty)$ for all $\theta \in \mathbb{R}$. Assume that $f_\theta(x)$ is twice continuously differentiable in the parameters θ, x. Show that $\{f_\theta\}$ has the MLR property with respect to the statistic $t(x) = x$ ($x \in \mathbb{R}$) if and only if $\frac{\partial^2}{\partial\theta\partial x} \log f_\theta(x) \geq 0, \forall x, \theta \in \mathbb{R}$.

Solution:

The Monotone Likelihood Ratio property for $t(x) = x$ holds if and only if the cross-partial derivative of the log-density is non-negative, which states that $\frac{\partial^2}{\partial\theta\partial x} \log f_\theta(x) \geq 0$. This relationship exists because the MLR property requires the likelihood ratio to grow with x, meaning the slope of the log-density must increase as the parameter θ increases.

We can prove this equivalence using a chain of standard calculus properties.

By definition, the family $\{f_\theta\}$ has the MLR property with respect to $t(x) = x$ if, for any $\theta_1 < \theta_2$, the likelihood ratio is non-decreasing in x:

$$\frac{f_{\theta_2}(x)}{f_{\theta_1}(x)} \text{ is non-decreasing in } x$$

Since $f_\theta(x) > 0$ and the natural logarithm is a strictly increasing function, this is strictly equivalent to:

$$\log\left(\frac{f_{\theta_2}(x)}{f_{\theta_1}(x)}\right) \text{ is non-decreasing in } x$$

$$\log f_{\theta_2}(x) - \log f_{\theta_1}(x) \text{ is non-decreasing in } x$$

A continuously differentiable function is non-decreasing if and only if its first derivative is non-negative. Taking the partial derivative with respect to x:

$$\frac{\partial}{\partial x}\left[\log f_{\theta_2}(x) - \log f_{\theta_1}(x)\right] \geq 0, \quad \forall x \in \mathbb{R}$$

$$\frac{\partial}{\partial x} \log f_{\theta_2}(x) \geq \frac{\partial}{\partial x} \log f_{\theta_1}(x), \quad \forall x \in \mathbb{R}, \text{ whenever } \theta_1 < \theta_2$$

The inequality above states that for any fixed x, the function $g(\theta) = \frac{\partial}{\partial x} \log f_\theta(x)$ preserves order when moving from θ_1 to θ_2.
This means that for every fixed $x \in \mathbb{R}$:

$$g(\theta) = \frac{\partial}{\partial x} \log f_\theta(x) \text{ is a non-decreasing function of } \theta$$

Since $f_\theta(x)$ is twice continuously differentiable, $g(\theta)$ is differentiable with respect to θ. A differentiable function is non-decreasing if and only if its derivative is non-negative:

$$\frac{\partial}{\partial\theta} g(\theta) \geq 0, \quad \forall x, \theta \in \mathbb{R}$$

$$\frac{\partial^2}{\partial\theta\partial x} \log f_\theta(x) \geq 0, \quad \forall x, \theta \in \mathbb{R}$$

Because each step in this logical chain uses an if and only if ($\iff$) property of differentiable functions, the proof is complete.

■

Exercise 9. Suppose X is a binomial distributed random variable with parameters $n = 100$ and $\theta \in [0, 1]$ where θ is unknown. Suppose we want to test the hypothesis H_0 that $\theta = 1/2$ versus the hypothesis H_1 that $\theta \neq 1/2$. Consider the hypothesis test that rejects the null hypothesis if and only if $|X - 50| > 10$.

Using e.g. the central limit theorem, do the following:

- Give an approximation to the significance level α of this hypothesis test

- Plot an approximation of the power function $\beta(\theta)$ as a function of θ.
- (Optional) Estimate p values for this test when $X = 50$, and also when $X = 70$ or $X = 90$.

Solution: To solve this problem, we model the binomial random variable $X \sim \text{Binomial}(n = 100, \theta)$ using the Central Limit Theorem. For a large sample size ($n = 100$), X is approximately normally distributed:

$$X \approx N(\mu, \sigma^2) = N\big(100\theta, 100\theta(1-\theta)\big)$$

The test rejects the null hypothesis $H_0 : \theta = 1/2$ if and only if $|X - 50| > 10$. This translates to a rejection region of $X < 40$ or $X > 60$.

The significance level α is the probability of rejecting H_0 when H_0 is true ($\theta = 1/2$). Under H_0:

- Mean: $\mu = 100(1/2) = 50$
- Standard Deviation: $\sigma = \sqrt{100 \times \frac{1}{2} \times \frac{1}{2}} = 5$

Using the CLT with a standard continuity correction (shifting the discrete boundaries by 0.5), the rejection region becomes $X \leq 39.5$ or $X \geq 60.5$:

$$z_1 = \frac{39.5 - 50}{5} = -2.10$$

$$z_2 = \frac{60.5 - 50}{5} = 2.10$$

$$\alpha = P(Z \leq -2.10) + P(Z \geq 2.10) = 2 \times \Phi(-2.10) \approx 0.0357 \text{ (or } 3.57\%)$$

The power function $\beta(\theta)$ represents the probability of rejecting H_0 given the true underlying parameter θ:

$$\beta(\theta) = P_\theta(X < 40) + P_\theta(X > 60)$$

Under any true parameter θ, the mean is $\mu = 100\theta$ and the standard deviation is $\sigma = \sqrt{100\theta(1-\theta)}$. Incorporating the continuity correction, the CLT approximation yields:

$$\beta(\theta) \approx \Phi\left(\frac{39.5 - 100\theta}{\sqrt{100\theta(1-\theta)}}\right) + \left(1 - \Phi\left(\frac{60.5 - 100\theta}{\sqrt{100\theta(1-\theta)}}\right)\right)$$

Below is the plotted approximation of this power function across the domain $\theta \in [0, 1]$.

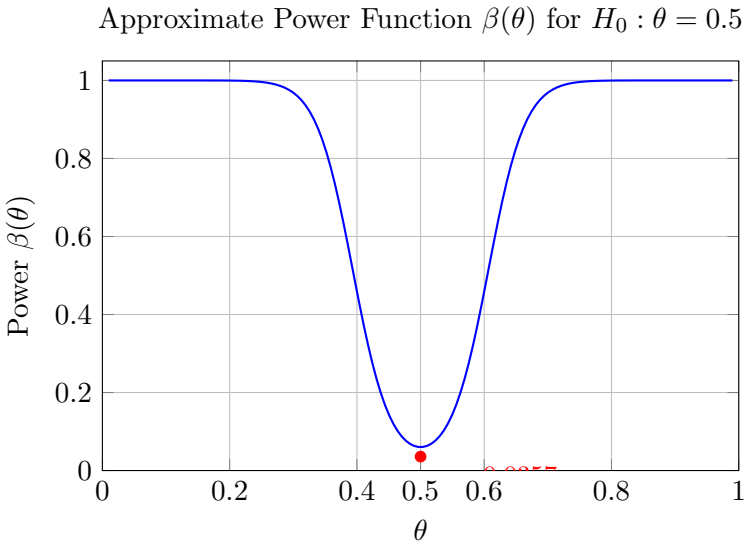


Figure 1: Power curve plotted using the Normal approximation with continuity correction.

The p-value is the probability of observing a test statistic at least as extreme as the one calculated, assuming H_0 is true ($\mu = 50, \sigma = 5$). Using the two-sided normal approximation with continuity correction:

- For $X = 50$:
  The observation lands exactly on the null mean. The probability of being at least as extreme as the mean encompasses the entire distribution.

$$\text{p-value} = 1.0000$$

- For $X = 70$:
  The distance from the mean is $|70 - 50| = 20$. Factoring in the continuity correction yields the critical threshold $X \geq 69.5$.

$$z = \frac{69.5 - 50}{5} = 3.90 \implies p - value = 2 \times \Phi(-3.90) \approx 0.0001$$

- For $X = 90$:
  The distance from the mean is $|90 - 50| = 40$. Factoring in the continuity correction yields the critical threshold $X \geq 89.5$.

$$z = \frac{89.5 - 50}{5} = 7.90 \implies \text{p-value} = 2 \times \Phi(-7.90) \approx 2.8 \times 10^{-15}$$

Exercise 10. Let $X_1, \ldots, X_n$ be a real-valued random sample of size n from a family of distributions $\{f_\theta : \theta \in \Theta\}$. Suppose $\Theta = \mathbb{R}$. Fix $\theta \in \mathbb{R}$. Denote $X := (X_1, \ldots, X_n)$. Consider a set of hypothesis tests $\phi_\alpha \colon \mathbb{R}^n \to [0, 1]$, for any $\alpha \in [0, 1]$. Assume that these tests are nested in the sense that $\phi_\alpha \leq \phi_{\alpha'}$ for all $0 \leq \alpha < \alpha' \leq 1$. Suppose we are testing the hypothesis H_0 that $\{\theta \leq \theta_0\}$ versus H_1 that $\{\theta > \theta_0\}$. Suppose also that $\{f_\theta\}$ has the monotone likelihood ratio property with respect to a statistic $Y = t(X)$ that is a continuous random variable.

- Show that the family of UMP tests with significance level at most α satisfies the nested property mentioned above (for all $\alpha \in [0, 1]$).
- (Optional) Show that, if $X = x$, then the p-value $p(x)$ satisfies

$$p(x) = P_{\theta_0}(t(X) > t(x)).$$

(A p-value is defined for randomized tests as $p(X) := \inf\{\alpha \in [0, 1] : \phi_\alpha(X) = 1\}$.)

Solution: By the Karlin-Rubin Theorem, since the family $\{f_\theta\}$ has the monotone likelihood ratio property in the continuous statistic $Y = t(X)$, the Uniformly Most Powerful level-α test for testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$ is given by:

$$\phi_\alpha(x) = \begin{cases} 1 & \text{if } t(x) > c_\alpha \\ 0 & \text{if } t(x) < c_\alpha \end{cases}$$

Because Y is a continuous random variable, the randomization at the boundary $t(x) = c_\alpha$ carries zero probability mass and can be ignored. The threshold c_α is uniquely determined by the size constraint at the boundary $\theta = \theta_0$:

$$P_{\theta_0}(t(X) > c_\alpha) = \alpha$$

Now, choose any significance levels $0 \leq \alpha < \alpha' \leq 1$. By definition:

- $P_{\theta_0}(t(X) > c_\alpha) = \alpha$
- $P_{\theta_0}(t(X) > c_{\alpha'}) = \alpha'$

Since $\alpha < \alpha'$, it follows that $P_{\theta_0}(t(X) > c_\alpha) < P_{\theta_0}(t(X) > c_{\alpha'})$. Because the survival function $g(c) = P_{\theta_0}(t(X) > c)$ is monotonically decreasing with respect to c, a larger probability requires a lower threshold:

$$c_\alpha > c_{\alpha'}$$

Let us evaluate the test functions for any given sample x:

- If $\phi_\alpha(x) = 1$, then $t(x) > c_\alpha$.
- Since $c_\alpha > c_{\alpha'}$, it transitively follows that $t(x) > c_{\alpha'}$.
- This immediately triggers $\phi_{\alpha'}(x) = 1$.

If $\phi_\alpha(x) = 0$, then $\phi_\alpha(x) \leq \phi_{\alpha'}(x)$ trivially holds because $\phi_{\alpha'}(x) \in [0, 1]$. Therefore, $\phi_\alpha(x) \leq \phi_{\alpha'}(x)$ for all $x \in \mathbb{R}^n$ and $0 \leq \alpha < \alpha' \leq 1$, which proves the nested property.

The p-value for a given observed sample $X = x$ is defined as the smallest level of significance α at which the null hypothesis is rejected:

$$p(x) = \inf\{\alpha \in [0, 1] : \phi_\alpha(x) = 1\}$$

From the structure of the non-randomized UMP test established above, the test rejects H_0 if and only if $t(x) > c_\alpha$. Let $g(c) = P_{\theta_0}(t(X) > c)$ represent the continuous survival function of $t(X)$ under θ_0. We rewrite the rejection condition as:

$$\phi_\alpha(x) = 1 \iff t(x) > c_\alpha$$

Applying the strictly decreasing function $g(\cdot)$ to both sides of the inequality reverses the direction:

$$t(x) > c_\alpha \iff g(t(x)) < g(c_\alpha)$$

Since $g(c_\alpha) = \alpha$ by definition, we substitute it into the expression:

$$\phi_\alpha(x) = 1 \iff g(t(x)) < \alpha$$

Thus, the set of all significance levels α that reject H_0 given $X = x$ is the interval $(g(t(x)), 1]$. Taking the infimum of this set yields:

$$p(x) = \inf\{\alpha \in (g(t(x)), 1]\} = g(t(x))$$

Substituting back the definition of $g(\cdot)$, we obtain the final result:

$$p(x) = P_{\theta_0}(t(X) > t(x))$$